INDIVIDUAL CONTRIBUTION REPORT | Member - 4

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Student ID | UOW Number	20240724 | W2181815
Group number	Group 05
Member Role	Student - 4
Module	3D Interactive Media Development
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GitHub	https://github.com/rashennimsara/TreasureIsland-GroupProject/branches

1. Introduction and project overview

Island Treasure Hunter is a group-developed 3D LEGO-style adventure set on a tropical island. The project combines an explorable environment, character movement and interactive obstacles to support a treasure-hunting experience. Individual members worked on separate systems before bringing their Unity assets into a shared repository. This report records my own contribution, the development choices behind it, collaboration, testing and areas that require further verification. It distinguishes prepared components from fully tested gameplay so that the evidence accurately reflects my work during CW1.

2. My role and responsibilities

My assigned responsibility was to prepare a cave-related trigger interaction and a breakable brick-wall system based on the Tutorial 04 concepts. I also worked with a LEGO test character so that the trigger could be checked with a player object instead of only inspecting the scene. My work included character setup, collision and movement investigations, a cave entrance trigger, separate stone pieces, wall explosion components and reusable prefabs. I contributed these assets to the group repository and participated in integration work. The trigger detection was observed in testing; the final wall-scattering result and complete gameplay integration must be confirmed separately. I also added interactive messages and a treasure chest interaction, supported player testing.

Assigned Area	**My contribution / status**
LEGO test player	Character model, Walk/Stand assets and movement checks
Cave entrance trigger	Box Collider set as trigger; Player-tag detection tested
Breakable stone wall	Individual stone pieces, colliders, rigidbodies and explosion setup
Reusable assets	Player, wall and trigger prefabs organised for integration

3. Development and implementation

3.1 LEGO character setup and movement checks

I imported a LEGO character model into Unity and placed it beneath a TestPlayer object to keep the visible character separate from the main player object. I worked with Walk and Stand model or animation assets and adjusted the character's position and scale to better match the island environment and the breakable wall. I investigated the Capsule Collider and Rigidbody components because movement and collision behaviour depend on how the player object is configured. I also worked on keyboard movement and jumping and tested the character in the island setting. During these checks, the relative scale of the player and wall was important: if the character appeared too large, the cave puzzle would not look believable. Character setup was supporting work for the trigger test, not a claim that I completed the group's final animated character controller.

3.2 Cave entrance trigger

I created a CaveEntranceTrigger GameObject near the proposed cave entrance. I added a Box Collider and enabled Is Trigger so that the area could detect an entering object without acting as a solid wall. I worked with CaveTrigger.cs and used the Player tag to identify the test character. When the player entered the trigger area, a Console message indicated that the trigger had detected the character. This was a useful initial test because it confirmed that the trigger volume, player tag and script event were

communicating. I adjusted the trigger's position to suit the planned cave entrance. I improved the trigger by adding an E-key interaction and a TextMeshPro message displaying "Press E to break the wall" in the Game view. When the player enters the trigger area and presses E, the wall explosion is activated.

3.3 Breakable stone wall and physics preparation

I imported a stone-wall model for testing and prepared a BreakableStoneWall object containing separate stone pieces, including Stone_01 through Stone_10. I worked with Box Colliders and Rigidbody components on the individual pieces so that they could participate in a physics-based scattering effect. I added or configured a WallExplosion script and connected the wall explosion component to the CaveEntranceTrigger setup. I also adjusted the wall's scale and placement in relation to the LEGO test character and the cave area. Separating the wall into pieces was necessary because one unbroken mesh would not behave like a brick array during an explosion. I improved the trigger by adding an E-key interaction and a TextMeshPro message displaying "Press E to break the wall" in the Game view. When the player enters the trigger area and presses E, the wall explosion is activated.

3.4 Prefabs and asset organisation

To make the interaction reusable, I created or organised prefabs for TestPlayer, BreakableStoneWall, CaveEntranceTrigger, CaveStoneWall and stone-wall. I also organised files under Member 4 folders for Models, Materials, Prefabs and Scripts. Prefabs are useful in a group project because they allow configured objects to be placed into another scene without rebuilding every component from the beginning. They can also help preserve the intended hierarchy of a player, trigger or wall. I checked how the assets were arranged before integration and considered whether script references would need to be assigned again in the combined scene. The purpose of this organisation was to reduce integration work and make it easier for teammates to identify which assets belonged to my contribution.

3.5 Treasure Chest Interaction and Animation

I added a treasure chest inside the cave and created TreasureChestTrigger.cs to detect the player. I used TextMeshPro to display "Press E to open chest" and a congratulations message. I also configured an Animator Controller with the ChestOpen state using the imported Take 001 animation clip for the chest-opening interaction

3.6 Explosion Force Slider

I investigated using a Unity UI Slider to adjust the explosion force of the breakable wall. The slider can be linked to the WallExplosion script through the On Value Changed event. This feature still requires final implementation and testing.

4. Collaboration, GitHub and integration

I used the shared TreasureIsland-GroupProject repository to exchange my assets and scripts with the group. The work record identifies a member4-triggers-4- branch and an integration-treasure-island branch. I committed and pushed Member 4 assets, and I worked with prefabs and Unity packages while combining project content. Scenes involved in the process included Member4_TriggerPuzzle_Test and TreasureIsland_Final_Combined2. Integrating Unity scenes was not always straightforward because different members could modify related files. I investigated GitHub merge conflicts and the need to preserve scene changes and .meta files. My contribution to integration was to make the player, trigger and breakable-wall setup available for group testing, rather than assuming that every interaction would automatically work when assets were merged.

5. Testing, challenges and solutions

The cave entrance trigger was tested by moving the Player-tagged test object into the trigger volume and observing a Console message. I also investigated several practical issues: character size compared with the wall, falling through the island ground, movement or jumping speed, unexpected rotation, possible collider interference and Input System compatibility. For the wall, I checked trigger placement and Rigidbody configuration when scattering behaviour was unexpected. These investigations helped identify which components could affect the puzzle, but the available notes do not confirm that every issue was fully resolved. The wall's final explosion, its timing and its behaviour in the combined island scene should be tested and documented with actual results before submission.

6. Individual reflection and future improvements

This work helped me understand the difference between trigger detection and a complete physics interaction. The Console message was useful evidence for the first step, while the wall required additional checks of separate pieces, Rigidbody settings and script connections. Preparing prefabs also showed me how reusable objects support group integration. I learned that character scale, ground colliders and input settings can affect apparently unrelated features. For CW2, I would focus on verifying a reliable trigger-to-wall sequence, refining the cave puzzle and testing it with the group's final player and camera. I would also keep clearer records of each successful and unsuccessful test.

7. AI assistance and responsible use

No	Question Asked to ChatGPT	AI Assistance Received

1	How can I press E to break the wall?	xplained how to detect the E key and activate the cave trigger.
2	Why is my wall not exploding?	Suggested checking Rigidbody components, colliders and script references.
3	How can I show a message in the Game view?	Explained how to use TextMeshPro to display “Press E to break the wall.”
4	How do I open a treasure chest using an animation?	Explained Animator Controllers, animation clips and the chest trigger.
5	Why is my player not jumping?	Suggested checking ground detection, Rigidbody and movement settings.
6	How can I add a slider to control explosion force?	Explained how a UI Slider could change the explosion force through its On Value Changed event.

8.References

Unity Technologies (n.d.) *Animator.Play*. Available at: https://docs.unity.com/en-us/engine/6000.0/script-reference/unityengine/animator/play (Accessed: 5 November 2026).

Unity Technologies (n.d.) *Creating text – TextMesh Pro*. Available at: https://docs.unity.cn/Packages/com.unity.ugui@2.0/manual/TextMeshPro/TMPObjects.html (Accessed: 10 November 2026).

Unity Technologies (n.d.) *OnTriggerEnter(Collider)*. Available at: https://docs.unity.com/en-us/engine/6000.5/script-reference/unityengine/monobehaviour/ontriggerenter (Accessed: 9 November 2026).

Tarodev (2021) Unity Slider in 4 Minutes – Unity Tutorial. Available at: https://www.youtube.com/watch?v=nTLgzvklgU8 (Accessed: 7 November 2026).

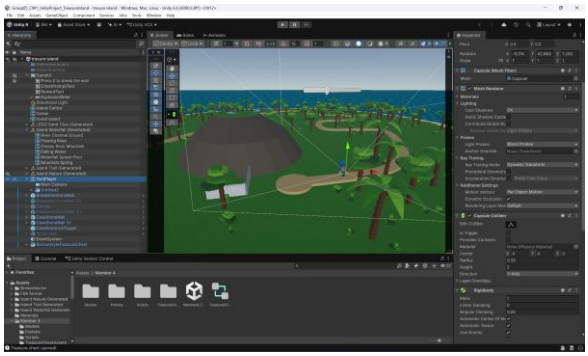
Unity Technologies (n.d.) Unity Asset Store. Available at: https://assetstore.unity.com/ (Accessed: 11 November 2026).

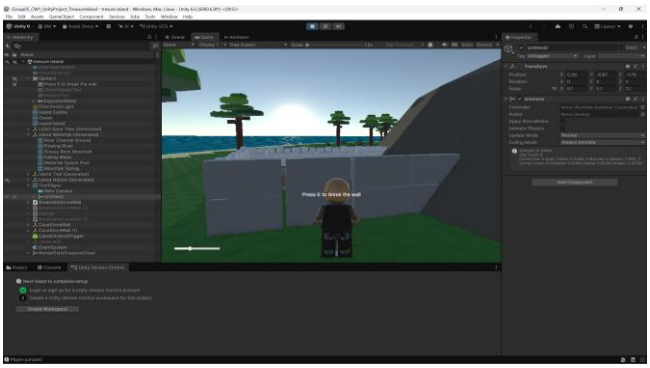
Bilal (2017) Unity Collision Tutorial – Colliders and Triggers. Available at: https://www.youtube.com/watch?v=6C4KfuW2q8Y (Accessed: 7 November 2026).

OpenAI (2026) ChatGPT [Large language model]. Available at: https://chatgpt.com/ (Accessed: 5 November 2026).

Appendix A — Evidence of individual contribution

Figure 1: TestPlayer hierarchy and collider / Rigidbody settings





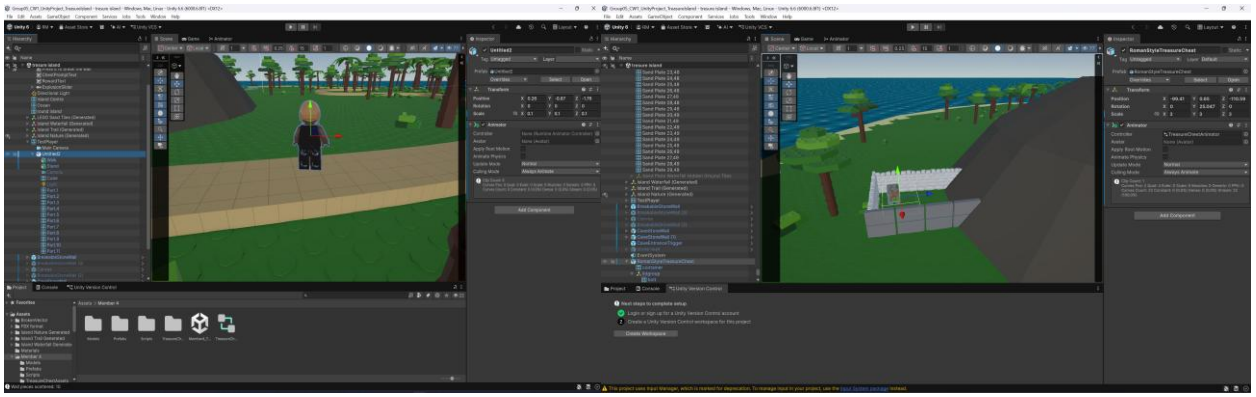
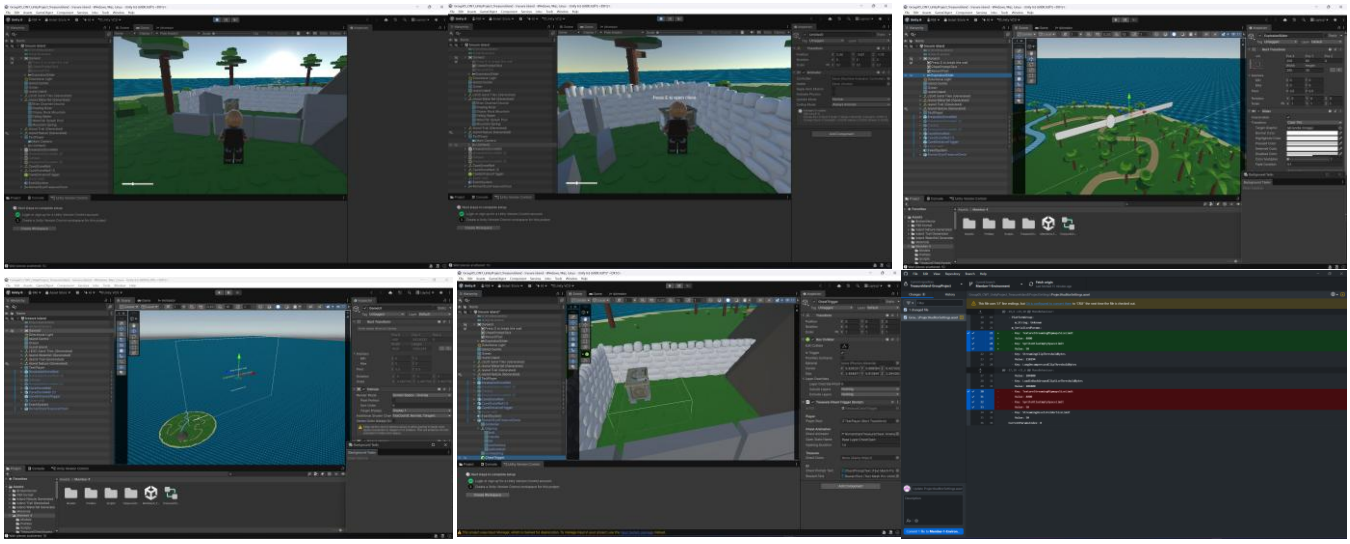


Figure 2: CaveEntranceTrigger with Box Collider and CaveTrigger script



Figure 3: BreakableStoneWall pieces, WallExplosion settings or GitHub commits



Appendix B — Additional supporting records

AI chat transcripts and AI summary

